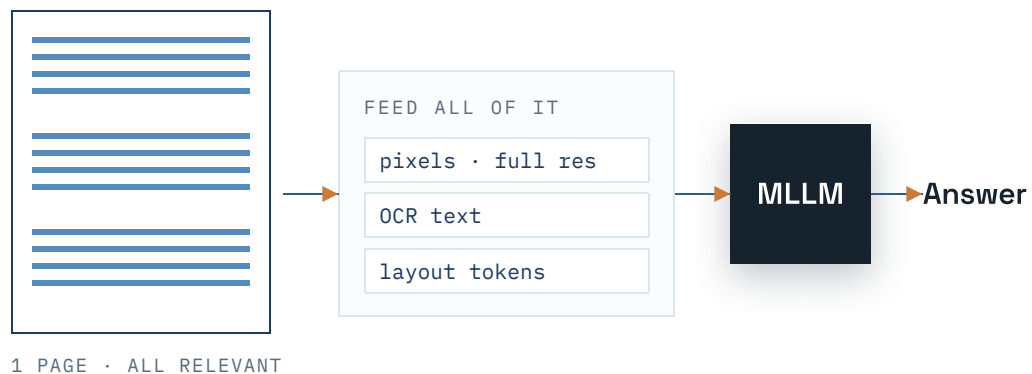


THE PROBLEM IS NOT A BIGGER VERSION OF THE OLD ONE

SP-VRDU vs. MP-VRDU

SP-VRDU

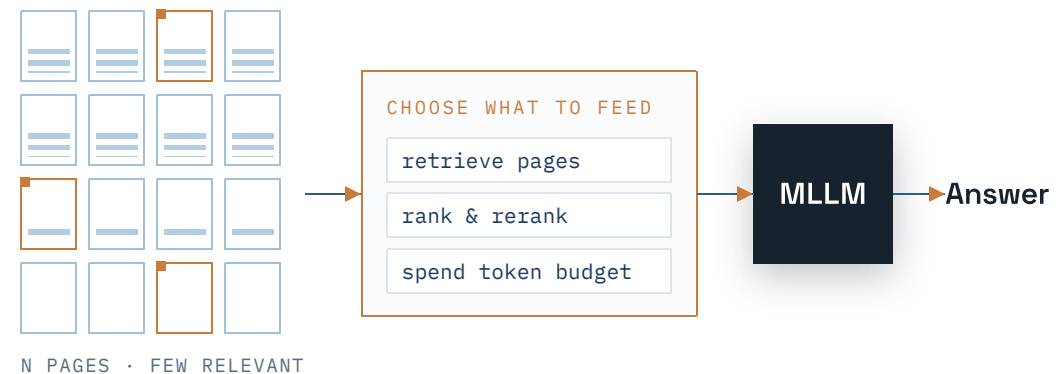
ONE PAGE



The page **fits**. Every channel is kept at full fidelity, so representation is not a choice.

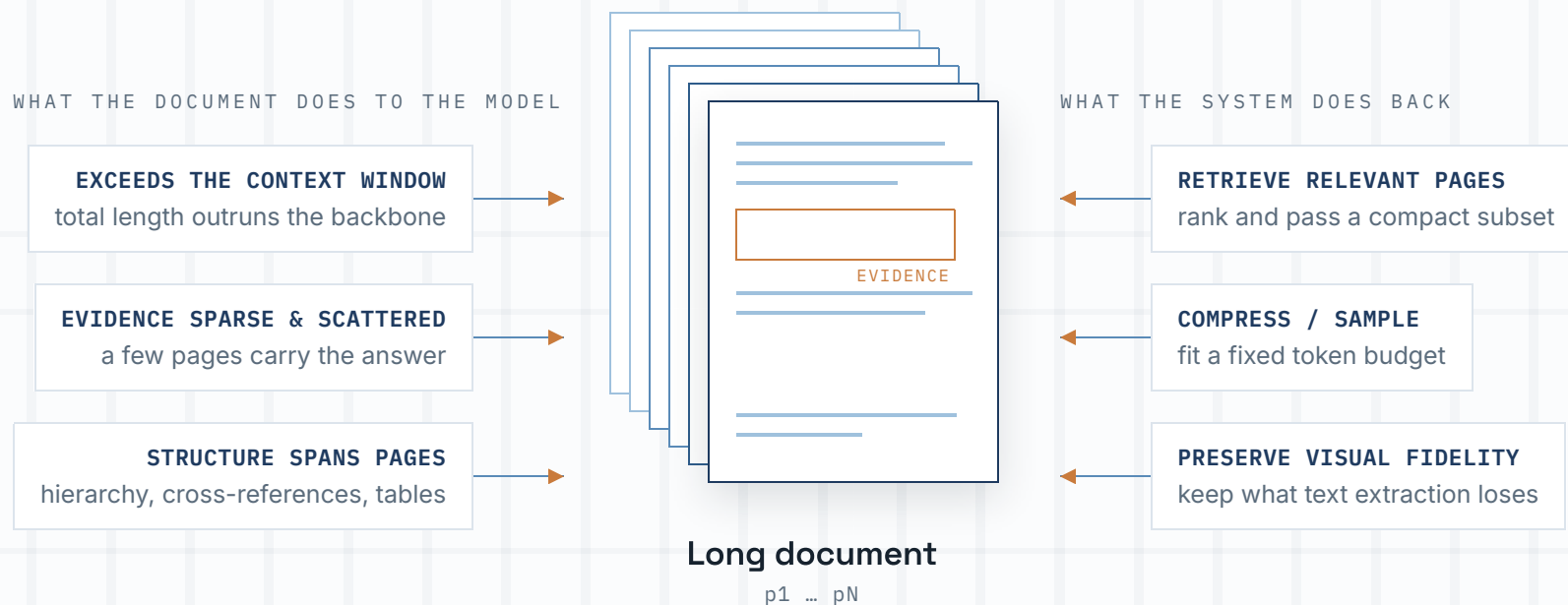
MP-VRDU

TENS-HUNDREDS OF PAGES



The document **does not fit**. The system must find the few pages that matter and read them at **enough** fidelity to answer.

An Evidence-Management Problem



The system must decide what to **retain**, **discard**, **retrieve**, or **acquire**.

What Makes Multi-Page Hard?

Every evidence decision is enacted over a **modality**. Three axes, three sets of pressures.

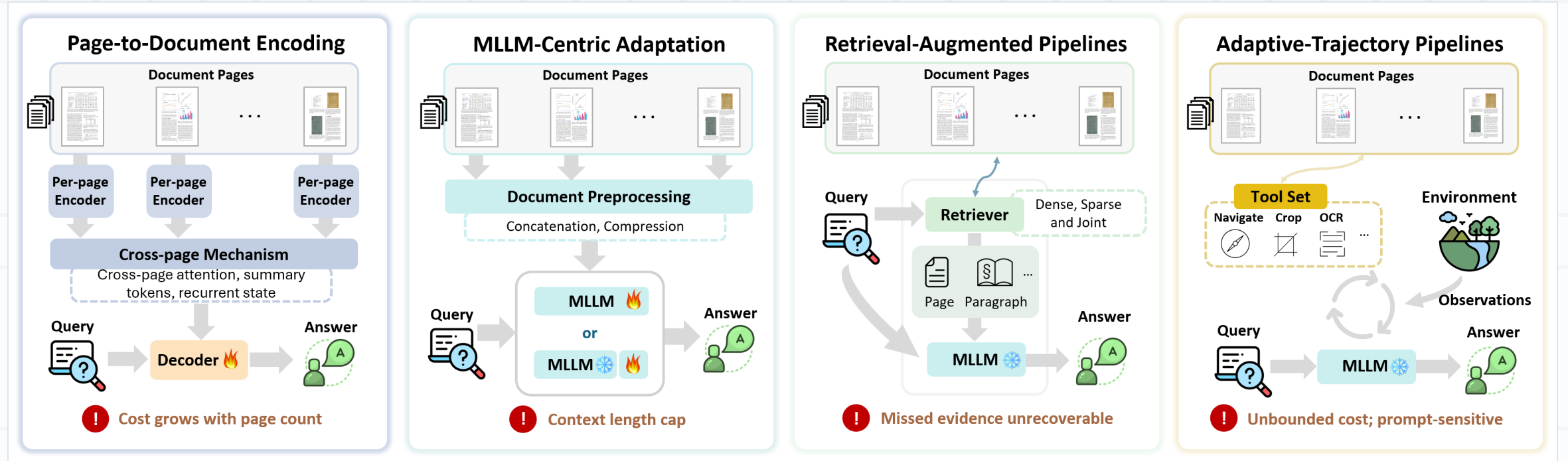
TEXT§4.1	VISUAL§4.2	STRUCTURE§4.3
Reading everything is impossible; every fix is lossy somewhere. ■ VOLUME length outruns the context ■ CONTINUITY paragraphs and tables break at page edges ■ NOISE OCR error accumulates across pages ■ HETEROGENEITY text density varies page to page	Pixels keep what text extraction loses, but they cost tokens. ■ RESOLUTION cannot hold full fidelity at document length ■ SIGNAL COMPETITION text, figures and layout share one budget ■ HETEROGENEITY visual complexity varies page to page	What turns isolated pages into one document. ■ HIERARCHY sections and reading order span the whole file ■ CROSS-PAGE REFERENCES “see Figure 3” points somewhere else ■ SPANNING ELEMENTS tables and paragraphs continue across pages

CROSS-CUTTING

Multi-hop reasoning: the answer is assembled from pages that are far apart, so no single page scores well on its own.

p3 → p7 → p12

MP-VRDU Architecture Overview



evidence handled **inside** the backbone — THE TREND — outside it, orchestrated at inference

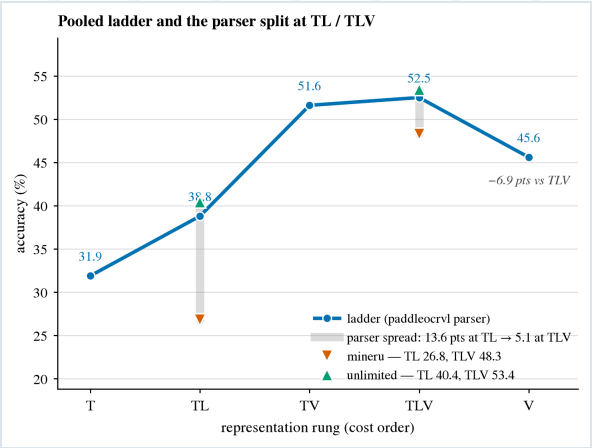
WHAT TO NOTICE

Each family pushes evidence handling further out of the backbone, through compression, then retrieval, then iterative orchestration. **None of them removes the bottleneck; each relocates it** (the red note under every panel is that family's price).

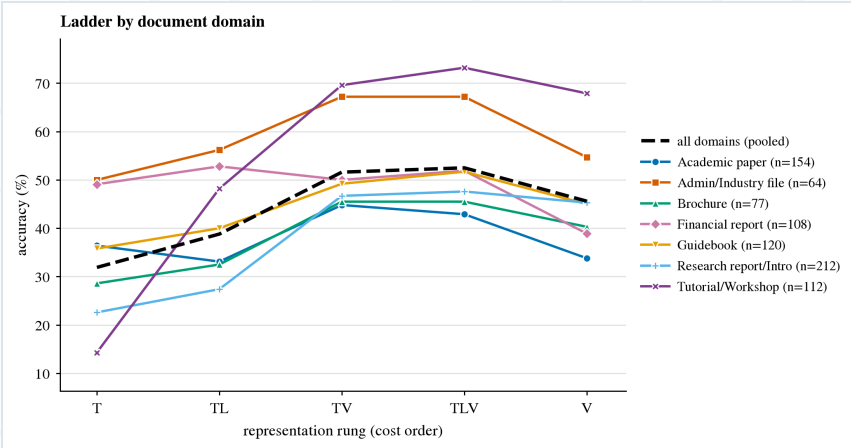
Accuracy vs. Evidence Richness



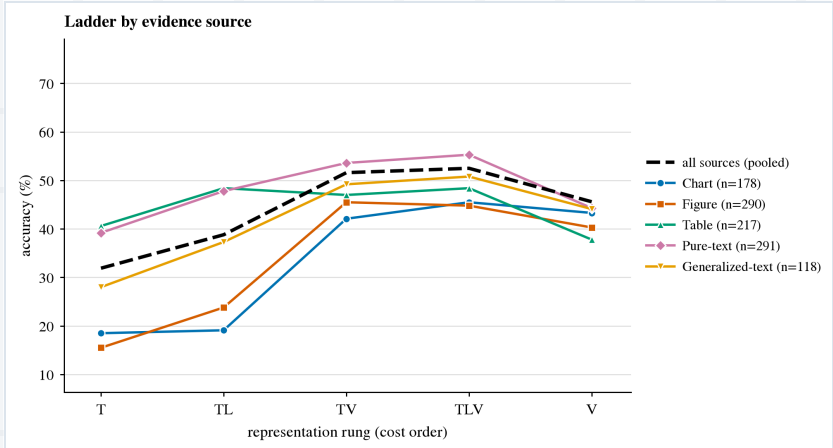
POOLED • ALL QUESTIONS



BY DOCUMENT DOMAIN • 7 DOMAINS



BY EVIDENCE SOURCE • 5 SOURCES



WHAT TO NOTICE

Accuracy climbs with richness and **peaks at TLV**, 31.9 to 52.5, but the ladder is not monotonic: dropping text for the page image alone costs 6.9 points. Richer rungs also **absorb parser quality**, with the spread between parsers falling from 13.6 points at TL to 5.1 at TLV. The shape holds across every domain and evidence source; only the height moves.

Missing vs. Distracting Evidence

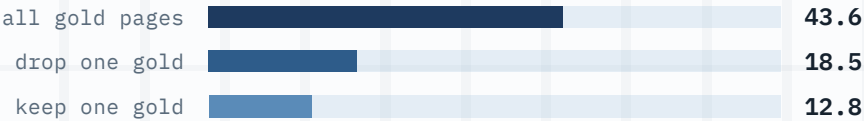
Page set	T	TL	TLV	V
Sufficiency — multi-hop pool, one gold page altered				
Oracle (all gold)	25.1	31.5	43.6	34.2
Drop bottom 1	11.4	13.4	18.5	15.1
Drop top 1	10.8	11.1	15.9	13.9
Keep bottom 1	9.4	9.7	10.5	9.9
Keep top 1	9.1	10.2	12.8	11.1
Robustness, 1-gold pool — add k distractors ($n=474$)				
Oracle ($k=0$)	37.6	45.1	64.6	55.7
+1 distractor	39.5	48.9	63.7	55.7
+2 distractors	38.8	47.9	64.3	56.1
+3 distractors	40.9	47.0	63.1	55.1
Robustness, 2-gold pool — add k distractors ($n\approx 241$)				
Oracle ($k=0$)	29.5	34.2	38.6	35.3
+1 distractor	25.7	34.2	38.6	34.4
+2 distractors	25.3	33.8	35.7	32.4
+3 distractors	21.6	28.3	33.6	33.2
Robustness, 3-gold pool — add k distractors ($n=40$)				
Oracle ($k=0$)	25.0	30.0	47.5	42.5
+1 distractor	22.5	40.0	45.0	40.0
+2 distractors	27.5	37.5	40.0	30.0
+3 distractors	25.0	35.0	40.0	32.5

Table 2: Selection under constructed page sets: sufficiency (leave-one-out) and robustness to distraction.

READING THE TABLE • TLV COLUMN

SUFFICIENCY

Losing evidence is unrecoverable



Removing a single required page costs **more than half the accuracy**. No rung of the ladder recovers it: the model answers anyway.

ROBUSTNESS • 1-GOLD POOL

Distraction is mostly absorbed



But tolerance **shrinks as more pages are required**, so the risk is not distraction itself, it is distraction on top of multi-hop evidence.



Single-Hop vs. Multi-Hop

COLUMNS

S single-hop

M multi-hop

Δ M minus S

ACROSS

T

TL

TLV

V

MULTI-HOP HERE IS MOSTLY TWO PAGES

Stratum	T			TL			TLV			V			n
	S	M	Δ	S	M	Δ	S	M	Δ	S	M	Δ	
By evidence source (overlapping)													
Chart	20.4	16.5	−4.0	19.4	19.0	−0.4	56.1	32.9	−23.2	54.1	30.4	−23.7	158
Figure	19.3	10.1	−9.2	25.3	21.8	−3.5	49.4	39.5	−9.9	45.8	33.6	−12.2	168
Generalized-text	30.4	26.7	−3.7	44.6	31.7	−13.0	66.1	38.3	−27.7	55.4	35.0	−20.4	116
Pure-text	46.8	31.3	−15.4	56.5	38.8	−17.7	67.5	42.5	−25.0	53.9	34.3	−19.6	274
Table	47.1	35.5	−11.7	66.3	32.7	−33.6	66.3	31.8	−34.5	51.9	25.5	−26.5	190
By domain (disjoint)													
Academic paper	34.5	39.7	+5.2	29.8	38.2	+8.5	39.3	47.1	+7.8	31.0	38.2	+7.3	152
Admin/Industry	58.5	40.0	−18.5	68.3	40.0	−28.3	87.8	35.0	−52.8	68.3	35.0	−33.3	61
Brochure	24.5	35.7	+11.2	30.6	35.7	+5.1	51.0	35.7	−15.3	44.9	32.1	−12.8	77
Financial report	60.3	34.1	−26.2	68.3	31.8	−36.4	74.6	20.5	−54.1	52.4	20.5	−31.9	107
Guidebook	42.5	25.5	−16.9	46.6	29.8	−16.8	60.3	38.3	−22.0	53.4	31.9	−21.5	120
Research report	28.0	17.1	−10.9	34.6	20.0	−14.6	64.5	30.5	−34.0	61.7	28.6	−33.1	212
Tutorial/Workshop	23.8	0.0	−23.8	54.0	41.3	−12.7	81.0	65.2	−15.7	81.0	52.2	−28.8	109
All	37.3	25.1	−12.2	45.0	31.3	−13.7	63.5	38.5	−25.0	55.2	33.5	−21.7	838

Table 3: The integration deficit (M−S) by evidence source and by domain across the ladder. The gap widens from T to TLV regardless of source or domain, locating the loss in cross-page integration rather than any single representation.

WHAT TO NOTICE

Richer input helps single-hop far more



gain from T to TLV

deficit −12.2 → −25.0

The gap **widens** up the ladder instead of closing, in every source and every domain.

Better representation does not buy integration. Joining the pages is the hard part, and almost every multi-hop question here is **just two pages**.